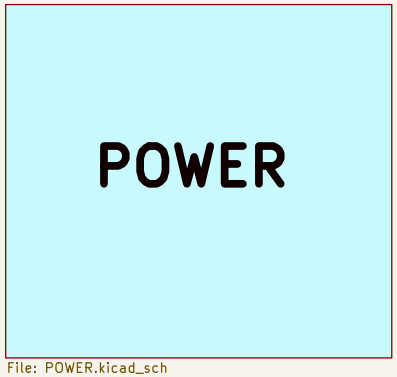
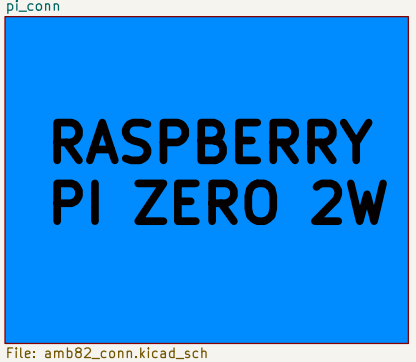


File: POWER.kicad_sch

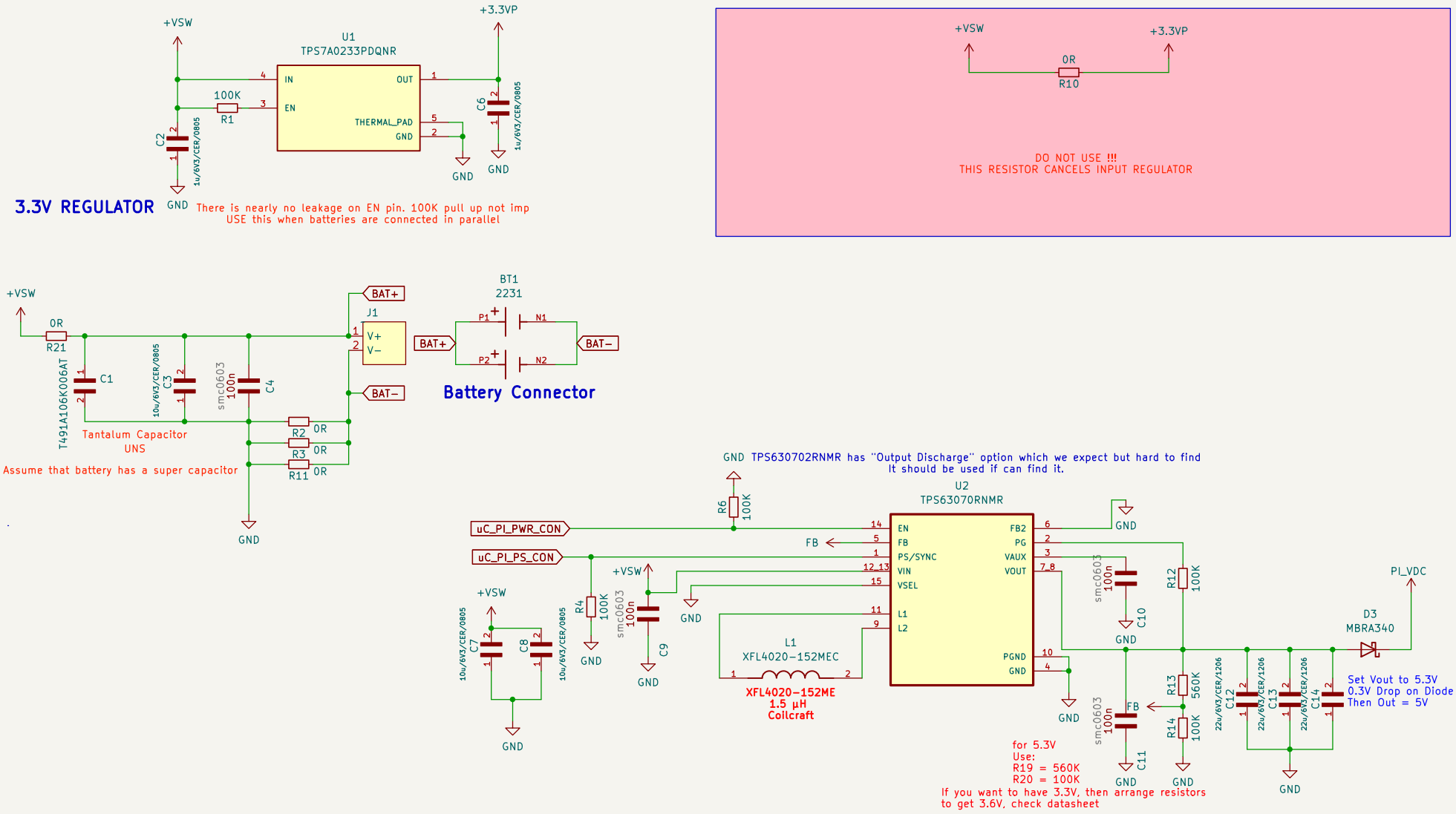


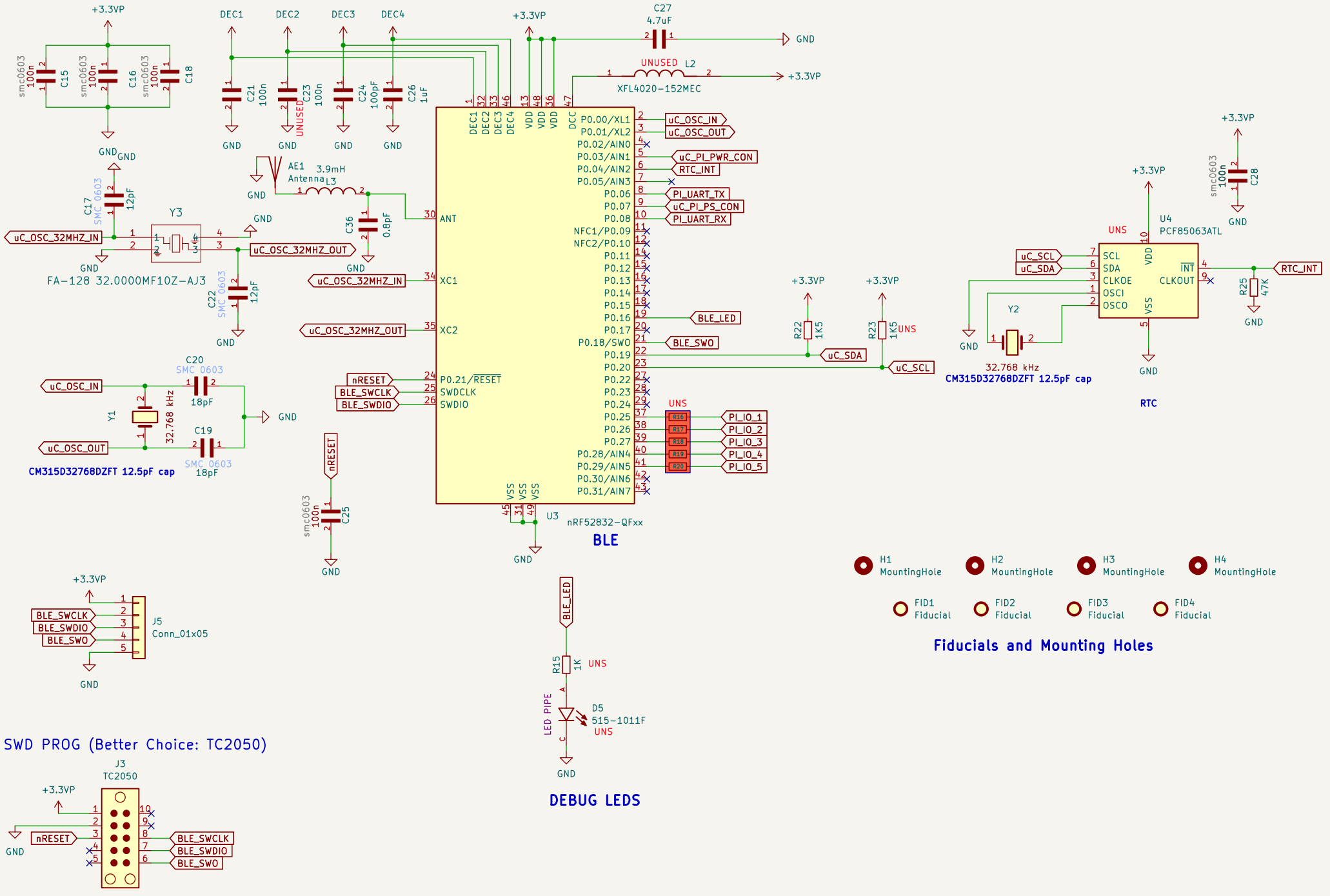
File: CPU.kicad_sch

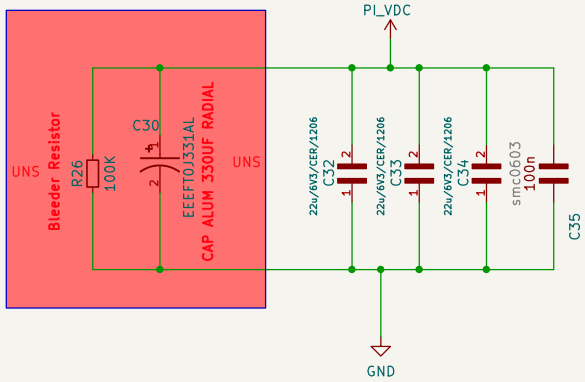
pi_conn

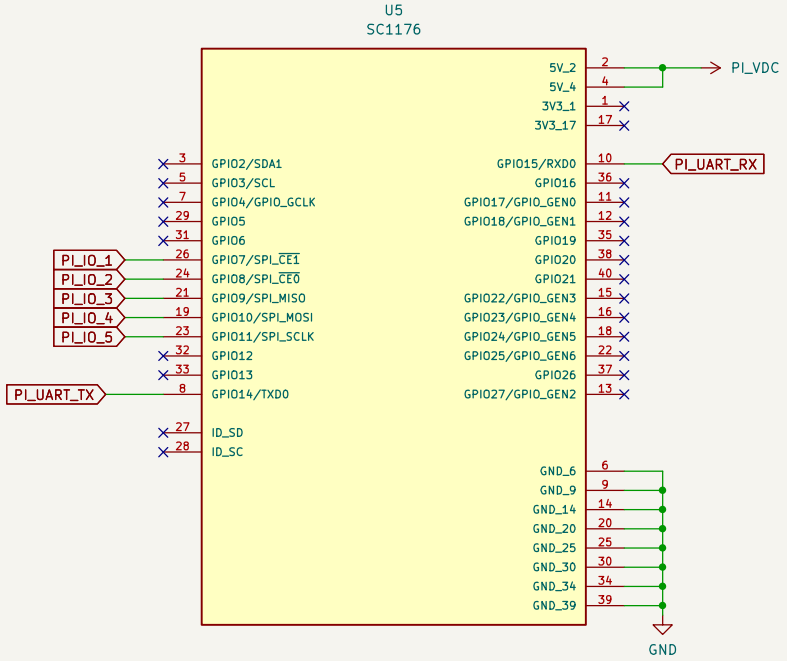


File: amb82_conn.kicad_sch


+VSW
U1
TPS7A0233PDQNR
+3.3VP
IN
OUT
EN
THERMAL_PAD
GND
100K
R1
C2
1u/6V3/CER/0805
C6
1u/6V3/CER/0805
GND
3.3V REGULATOR
There is nearly no leakage on EN pin. 100K pull up not imp
USE this when batteries are connected in parallel
+VSW
0R
R10
+3.3VP
DO NOT USE !!!
THIS RESISTOR CANCELS INPUT REGULATOR
+VSW
0R
R21
C1
T491A106K006AT
Tantalum Capacitor
UNS
Assume that battery has a super capacitor
C3
10u/6V3/CER/0805
C4
smc0603
100n
R2
0R
R3
0R
R11
0R
GND
BAT+
J1
V+
V-
BAT-
BT1
2231
P1
N1
P2
N2
BAT+
BAT-
Battery Connector
GND TPS630702RNMR has "Output Discharge" option which we expect but hard to find
It should be used if can find it.
U2
TPS63070RNMR
R6
100K
uC_PI_PWR_CON
uC_PI_PS_CON
FB
EN
FB
PS/SYNC
VIN
VSEL
L1
L2
PGND
GND
FB2
PG
VAUX
VOUT
+VSW
C7
10u/6V3/CER/0805
C8
10u/6V3/CER/0805
R4
100K
C9
smc0603
100n
L1
XFL4020-152MEC
XFL4020-152ME
1.5 µH
Coilcraft
C10
smc0603
100n
R12
100K
PI_VDC
D3
MBRA340
R13
560K
R14
100K
FB
C11
smc0603
100n
C12
22u/6V3/CER/1206
C13
22u/6V3/CER/1206
C14
22u/6V3/CER/1206
Set Vout to 5.3V
0.3V Drop on Diode
Then Out = 5V
for 5.3V
Use:
R19 = 560K
R20 = 100K
If you want to have 3.3V, then arrange resistors
to get 3.6V, check datasheet


+3.3VP
C15
C16
C18
DEC1
DEC2
DEC3
DEC4
C21 100n
C23 100n
C24 100pF
C26 1uF
C27 4.7uF
GND
UNUSED L2
XFL4020-152MEC
AE1 Antenna
3.9mH L3
C17 12pF
Y3
uC_OSC_32MHZ_IN
uC_OSC_32MHZ_OUT
FA-128 32.0000MF10Z-AJ3
C22 12pF
C36 0.8pF
ANT
XC1
XC2
C20 18pF
C19 18pF
Y1 32.768 kHz
CM315D32768DZFT 12.5pF cap
nRESET
BLE_SWCLK
BLE_SWDIO
P0.21/RESET
SWDCLK
SWDIO
C25
U3 nRF52832-QFxx
BLE
P0.00/XL1
P0.01/XL2
P0.02/AIN0
P0.03/AIN1
P0.04/AIN2
P0.05/AIN3
P0.06
P0.07
P0.08
NFC1/P0.09
NFC2/P0.10
P0.11
P0.12
P0.13
P0.14
P0.15
P0.16
P0.17
P0.18/SWO
P0.19
P0.20
P0.22
P0.23
P0.24
P0.25
P0.26
P0.27
P0.28/AIN4
P0.29/AIN5
P0.30/AIN6
P0.31/AIN7
uC_OSC_IN
uC_OSC_OUT
uC_PI_PWR_CON
RTC_INT
PI_UART_TX
uC_PI_PS_CON
PI_UART_RX
BLE_LED
BLE_SWO
uC_SDA
uC_SCL
PI_IO_1
PI_IO_2
PI_IO_3
PI_IO_4
PI_IO_5
R22 1K5
R23 1K5
UNS
U4 PCF85063ATL
SCL
SDA
CLKOE
OSCI
OSCO
CLKOUT
INT
VDD
VSS
Y2 32.768 kHz
CM315D32768DZFT 12.5pF cap
R25 47K
RTC
J5 Conn_01x05
SWD PROG (Better Choice: TC2050)
J3 TC2050
R15 1K
D5 515-1011F
LED PIPE
DEBUG LEDS
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
FID1 Fiducial
FID2 Fiducial
FID3 Fiducial
FID4 Fiducial
Fiducials and Mounting Holes





Use a paralel resistance to charge and discharge capacitors safely
!!
Observe transient with an oscilloscope
and set R and C to get a proper time constant.
MAY BE EEUFS0J472B and a Bleeder Resistor
!!

If we use TPS630702RNMR then we will not need that resistor as it has
Quick Output Discharge (QOD) ability